

# Project Portfolio Summary

skillsYNC | a skillit company

Pair B | Abdul Ghafoor · Ahmed Qureshi

AI/ML Engineering Track

## 01 The Ledger — Distributor Credit Risk Engine ■ skillsYNC Problem 01

*A credit scorecard that reads a distributor's own payment ledger and returns a defensible 300–900 risk score for every dealer.*

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product         | Web application (FastAPI + Next.js) that ingests three raw CSV exports — dealers, salesmen, invoice history — and returns a ranked risk table, per-dealer reason codes, and a downloadable Word risk memo for each account. Deployed and publicly accessible.                                                                                                                                                                                                                                        |
| Target audience | Owners of Pakistani FMCG, pharmaceutical and textile distribution houses managing 50–500 dealers, together with the finance leads and 3–10 field salesmen operating under them. Deliberately built for businesses with <i>no</i> ERP and no data team.                                                                                                                                                                                                                                               |
| Problem         | Trade credit is extended on a salesman's word — “ <i>yaar, acha banda hai</i> ” — with no structured assessment. Distributors report 3–7% of revenue lost to bad debt, PKR 1M+ in annual write-offs, and 100+ hours a year chasing payments. The salesman who vouches carries none of the downside risk.                                                                                                                                                                                             |
| How it works    | Six locally-grounded behavioural signals are engineered from raw invoice history: post-dated cheque bounce rate, Eid/Ramzan-adjusted lateness, salesman-vouch bias (leave-one-out), territory clustering, PKR-inflation-adjusted exposure, and order-frequency trend. A Weight-of-Evidence logistic scorecard converts these to a 300–900 score with plain-language reason codes. Features and outcomes are split temporally, so the model predicts future behaviour rather than restating the past. |

### ADVANTAGE OVER ALTERNATIVES

**vs. salesman judgement:** surfaces risk in accounts the sales team personally vouches for — five such dealers were flagged in the reference portfolio. **vs. credit bureaus:** works where no bureau coverage exists for small retailers. **vs. black-box ML:** every score decomposes into named, auditable reasons an owner can challenge. **vs. generic scorecards:** adapts to each uploaded portfolio — reads unfamiliar column names and date formats, derives its own feature window and market-specific late-payment threshold, and trains a portfolio-specific model when the general one is a poor fit. **Uniquely, it reports its own reliability** and declines to score rather than producing confident output on unsuitable data.

**Limitation:** validated on constructed portfolios with known outcomes; no real-world default has been observed yet. The RED/AMBER/GREEN tier is the reliable signal, the exact score directional.

## 02 Law Firm RAG System — Legal Research Assistant ■ skillsYNC Problem 07

*A retrieval-augmented assistant that answers questions from a firm's own case files and drafts legal memos with traceable citations.*

|                 |                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product         | A Retrieval-Augmented Generation pipeline over a law firm's private document corpus, served through a React interface. Supports natural-language querying of case files and automated generation of structured legal memos grounded in retrieved source passages.                                                                                                               |
| Target audience | Small and mid-sized Pakistani law firms — partners, associates and paralegals — who hold years of case history in unindexed folders of PDFs and scanned documents.                                                                                                                                                                                                              |
| Problem         | Legal research inside a firm's own archive is manual, slow and dependent on whoever remembers the file. Junior associates spend hours locating precedent that already exists in-house, and knowledge leaves the firm when a lawyer does.                                                                                                                                        |
| How it works    | Documents are chunked and embedded using multilingual embeddings suited to English/Urdu legal text, then indexed in a ChromaDB vector store. A router query engine directs each question to the appropriate retrieval strategy, and the generation layer answers strictly from retrieved context — with citations back to the source document, so no assertion is unverifiable. |

### ADVANTAGE OVER ALTERNATIVES

**vs. manual search:** reduces precedent-finding from hours to seconds across the firm's entire archive. **vs. general-purpose chatbots:** answers only from the firm's own documents, with citations — eliminating the fabricated-case-law risk that has already drawn sanctions in foreign courts. **vs. foreign legal platforms:** handles Pakistani legal material and mixed English/Urdu text, which international products do not cover. **vs. cloud legal SaaS:** the corpus stays under the firm's control, preserving client confidentiality and privilege.

**Limitation:** answer quality is bounded by the quality and coverage of the firm's own archive; the system retrieves and cites, it does not exercise legal judgement.

Both projects target Pakistani small and mid-sized businesses underserved by international software: sectors where the data is real but unstructured, and where an explainable, locally-grounded system is worth more than a technically sophisticated opaque one.